SOC-5811 Week 10: Panel Data

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Panel data

- What is “panel data?”

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- **Panel data:** A data set where we observe the same units (e.g., individuals, firms, countries, schools) over more than one time period.

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- **Panel data:** A data set where we observe the same units (e.g., individuals, firms, countries, schools) over more than one time period.
- Walk through data examples: when is longitudinal data panel data? Repeated observations of the same variable.

Basic linear regression model:

$$y_i = \beta_0 + \beta_1 X_i$$

Linear regression model with panel data:

$$y_{i,t} = \beta_0 + \beta_1 X_{i,t}$$

Panel data

- Let's walk through an example with data: traffic deaths and beer taxes by state/year.

Panel data

```r
dim(Fatalities)
```

```
## [1] 336  34
```

```r
Fatalities %>% select(state) %>% table()
```

```
## state
## al az ar ca co ct de fl ga id il in ia ks ky la me md ma mi mn ms mo mt ne nv
##  7  7  7  7  7  7  7  7  7  7  7  7  7  7  7  7  7  7  7  7  7  7  7  7  7  7
## nh nj nm ny nc nd oh ok or pa ri sc sd tn tx ut vt va wa wv wi wy
##  7  7  7  7  7  7  7  7  7  7  7  7  7  7  7  7  7  7  7  7  7  7
```

```r
Fatalities %>% select(year) %>% table()
```

```
## year
## 1982 1983 1984 1985 1986 1987 1988
##   48   48   48   48   48   48   48
```

Panel data

- This is called “balanced” panel data. Why? What would be “unbalanced” panel data?

Panel data

- This is called "balanced" panel data. Why? What would be "unbalanced" panel data?
- What shape is this dataset with respect to time?

Panel data

```r
head(Fatalities)
```

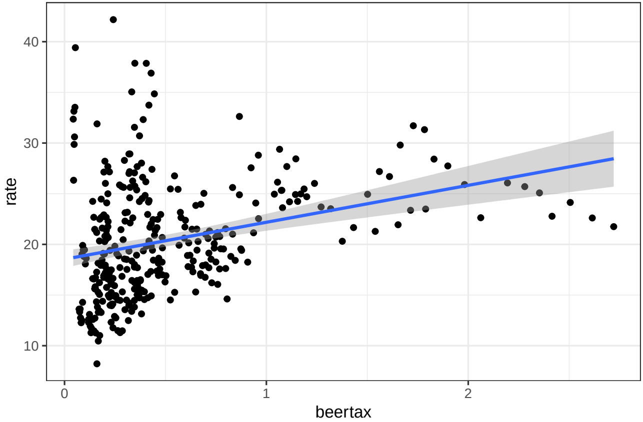
```
## # A tibble: 6 x 34
##   state year  spirits unemp income emppop beertax baptist mormon drinkage   dry
##   <fct> <fct>   <dbl> <dbl>  <dbl>  <dbl>   <dbl>   <dbl>  <dbl>    <dbl> <dbl>
## 1 al    1982     1.37 14.4  10544.   50.7    1.54    30.4  0.328     19    25.0
## 2 al    1983     1.36 13.7  10733.   52.1    1.79    30.3  0.343     19    23.0
## 3 al    1984     1.32 11.1  11109.   54.2    1.71    30.3  0.359     19    24.0
## 4 al    1985     1.28  8.90 11333.   55.3    1.65    30.3  0.376     19.7  23.6
## 5 al    1986     1.23  9.80 11662.   56.5    1.61    30.3  0.393     21    23.5
## 6 al    1987     1.18  7.80 11944    57.5    1.56    30.2  0.411     21    23.8
## # i 23 more variables: youngdrivers <dbl>, miles <dbl>, breath <fct>,
## #   jail <fct>, service <fct>, fatal <int>, nfatal <int>, sfatal <int>,
## #   fatal1517 <int>, nfatal1517 <int>, fatal1820 <int>, nfatal1820 <int>,
## #   fatal2124 <int>, nfatal2124 <int>, afatal <dbl>, pop <dbl>, pop1517 <dbl>,
## #   pop1820 <dbl>, pop2124 <dbl>, milestot <dbl>, unempus <dbl>,
## #   emppopus <dbl>, gsp <dbl>
```

Panel data

$$\text{Rate}_{i,t} = \beta_0 + \beta_1 \text{Tax}_{i,t}$$

PANEL DATA

All years (pooled model):



Panel data

```r
mod <- lm(rate ~ beertax, data=Fatalities)
summary(mod)
```

```
##
## Call:
## lm(formula = rate ~ beertax, data = Fatalities)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.9060  -3.7768  -0.9436   2.8548  22.7643
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  18.5331     0.4357  42.539  < 2e-16 ***
## beertax       3.6461     0.6217   5.865 1.08e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.437 on 334 degrees of freedom
## Multiple R-squared:  0.09336,    Adjusted R-squared:  0.09065
## F-statistic: 34.39 on 1 and 334 DF,  p-value: 1.082e-08
```

We could just think of panel data as several samples and several regression models:

$$\text{Rate}_{i,1988} = \beta_0 + \beta_1 \text{Tax}_{i,1988}$$

$$\text{Rate}_{i,1982} = \beta_0 + \beta_1 \text{Tax}_{i,1982}$$

Panel data

$$\text{Rate}_{i,1988} = \beta_0 + \beta_1 \text{Tax}_{i,1988}$$

$$\text{Rate}_{i,1982} = \beta_0 + \beta_1 \text{Tax}_{i,1982}$$

$$\text{Rate}_{i,1988} - \text{Rate}_{i,1982} = \beta_0 + \beta_1 (\text{Tax}_{i,1988} - \text{Tax}_{i,1982})$$

We have now made this a "within" estimator: we are predicting within-unit changes.

Panel data

```r
Fatalities_wide <- Fatalities %>%
  filter(year %in% c(1982,1988)) %>%
  pivot_wider(id_cols = state,
              values_from = c(rate,beertax),
              names_from = year) %>%
  mutate(rate_change = rate_1988 - rate_1982,
         tax_change = beertax_1988 - beertax_1982)
```

Panel data

```r
head(Fatalities_wide)
```

```
## # A tibble: 6 x 7
##   state rate_1982 rate_1988 beertax_1982 beertax_1988 rate_change tax_change
##   <fct>     <dbl>     <dbl>        <dbl>        <dbl>       <dbl>      <dbl>
## 1 al         21.3      24.9        1.54         1.50         3.66    -0.0379
## 2 az         25.0      27.1        0.215        0.346        2.07     0.132
## 3 ar         23.8      25.5        0.650        0.525        1.63    -0.126
## 4 ca         18.6      19.0        0.107        0.0866       0.417   -0.0208
## 5 co         21.7      15.1        0.215        0.173       -6.69    -0.0416
## 6 ct         16.5      15.0        0.224        0.217       -1.50    -0.00713
```

Panel data

```r
mod <- lm(rate_change ~ tax_change, data=Fatalities_wide)
summary(mod)
```

```
##
## Call:
## lm(formula = rate_change ~ tax_change, data = Fatalities_wide)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.2715  -0.9619   0.9212   2.2290   6.7745
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.7204     0.6064  -1.188   0.2410
## tax_change  -10.4097     4.1723  -2.495   0.0162 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.94 on 46 degrees of freedom
## Multiple R-squared:  0.1192, Adjusted R-squared:    0.1
## F-statistic: 6.225 on 1 and 46 DF,  p-value: 0.01625
```

Panel data

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- There was some variable we were missing. If we looked *within* penguin species, we found the opposite relationship!

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- Simpson's Paradox! What was the idea of Simpson's Paradox related to omitted variables?
- There was some variable we were missing. If we looked *within* penguin species, we found the opposite relationship!
- **Intercept shifts.**

- What is the difference between Simpson's Paradox examples and what we are doing here?

Causal inference

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Causal inference

- What is the difference between Simpson's Paradox examples and what we are doing here?
- There is no new *observed variable* that we are adding.
- So what have we done? We have accounted for *unobserved variables*.
- But what specific kind of unobserved variables? Time-invariant

Causal inference

- Panel data and causal inference

$$\text{Rate}_{i,1988} = \beta_0 + \beta_1 \text{Tax}_{i,1988} + \beta_2 Z_i$$

$$\text{Rate}_{i,1982} = \beta_0 + \beta_1 \text{Tax}_{i,1982} + \beta_2 Z_i$$

$$\text{Rate}_{i,1988} - \text{Rate}_{i,1982} = \beta_0 + \beta_1 (\text{Tax}_{i,1988} - \text{Tax}_{i,1982})$$

- When might we run into problems?